

Krishnaa Vadivel

Scientific Software Engineer | GPU and Systems

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Portfolio: <https://krishnaa423.github.io/personal-website/> | GitHub: <https://github.com/krishnaa423>

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Summary

Scientific-software engineer with C++/Python, GPU, embedded, database, and research-computing experience. Builds numerical and data-intensive tools from distributed simulation kernels through engineering-facing applications.

Technical Skills

Languages: Python, C++, C, C#, SQL, CUDA, OpenCL, OpenGL, Bash

Scientific/software: MPI, OpenMP, PETSc, SLEPc, DOLFINx, CMake, NumPy, SciPy, PyTorch, JAX

Systems: Linux, Docker, Kubernetes, ASP.NET, Microsoft SQL Server, Git, TotalView, Linaro DDT

Experience

PhD Researcher

Yale University, Electrical Engineering

2021–present

- Implement first-principles and many-body simulation capabilities in Python/C++ with MPI, OpenMP, PETSc, and SLEPc on Frontier and Perlmutter.
- Build ML-assisted scientific workflows with equivariant GNNs, JAX/PyTorch, and automatic differentiation for materials-property calculations.
- Package reproducible scientific environments in Docker and diagnose distributed failures with scaling runs and parallel debuggers.

Software and Embedded Systems Engineer

Probe Technologies / Weatherford Wireline Services, Houston, TX

2019–2021

- Shipped CUDA/OpenCL waveform-processing and OpenGL visualization applications for deployed downhole-tool diagnostics.
- Implemented C# desktop tools and ASP.NET/SQL Server services for customer input, remote test control, sensor calibration, and data collection.
- Collaborated on firmware, FPGA, board-debugging, and instrumentation workflows when software behavior depended on field hardware.

Selected Projects

Scientific workflow automation — Created retrieval-driven tooling that turns papers and documentation into structured starter inputs for simulation software.

Finite-element suite — Implemented and visualized heat, Poisson, and flow simulations in DOLFINx/PETSc.

Molecular VQE — Connected PySCF electronic-structure data to Qiskit Hamiltonian construction, UCCSD VQE, and exact references.

Education

PhD, Electrical Engineering

Yale University

2021–present

MPhil and MS, Electrical Engineering

Yale University

2023

MEng and BS, Electrical and Computer Engineering

Cornell University,

2013–2018

Research Record

Co-author, *Journal of the American Chemical Society*, 2025; APS presentations, 2024–2026.